ECE316 Formula

Euler Indentities

$$e^{j\theta} = \cos\theta + j\sin\theta, \quad \cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \quad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Average Power & Energy

Average Power of a signal $x(t)$ is defined as

$$P_x = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \left|x^2(t)\right| dt$$

The energy of $x(t)$ is

$$E_x = \lim_{T\to\infty} \int_{-T}^{T} \left|x^2(t)\right| dt = \int_{-\infty}^{\infty} \left|x^2(t)\right| dt$$

- $x(t)$ is an energy signal iff $0 \leq E < \infty$; i.e. $x(t)$ has finite energy. (iff means if and only if)
- $x(t)$ is an power signal iff $0 < P < \infty$; i.e. $x(t)$ has finite power. Note that $E = \infty$ for power signals.

Sinusoidal Signal

$$x(t) = Acos(2\pi f_c t + \theta)$$

where A is amplitude, f_c is frequency (in Hz), and θ is phase

Fourier Series

$$x(t) = \sum_{n=-\infty}^{\infty} X_n \phi_n(t) = \sum_{n=-\infty}^{\infty} X_n e^{j2\pi n f_0 t}$$

where $f_0 = \frac{1}{T}$.

$$\int_{T_0} x(t) e^{-j2\pi k f_0 t} dt = T_0 X_k$$

Derivation:

$$\text{Both Sides} \times e^{-j2\pi k f_0 t}$$

CTFS (from ECE216)

$$x(t) = \sum_{k=-\infty}^{\infty} \alpha_k e^{j\omega_0 k t}, \quad \alpha_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-j\omega_0 k t} dt, \quad \frac{1}{T_0} \|x\|^2 = \sum_{k=-\infty}^{\infty} |\alpha_k|^2$$

Angle Sum and Difference Identity

$$\cos(A+B) = \cos(A)\cos(B) \mp \sin(A)\sin(B)$$
$$\sin(A+B) = \sin(A)\cos(B) \pm \cos(A)\sin(B)$$

Sinc Function

$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

Note that sinc *function have different definitions, in the course we use this definition.*

Parseval's Theorem

$$P_x = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |X_K|^2$$

Rayleigh Energy Theorem

Rayleigh Energy Theorem is a continuous version of Parseval Theorem.

The **energy** of a signal $g(t)$ is given by:

$$E = \int_{-\infty}^{\infty} |g(t)|^2 \, dt = \int_{-\infty}^{\infty} |G(f)|^2 \, df$$

(in Joules/Hz)

Let:

$$\psi_g(f) = |G(f)|^2$$

Then $\psi_g(f)$ is called the **energy spectral density**, because the energy of $g(t)$ equals:

$$E = \int_{-\infty}^{\infty} \psi_g(f) \, df$$

Fourier Transform

Fourier Transform

$$G(f) = \int_{-\infty}^{\infty} g(\lambda) e^{-j2\pi f\lambda} d\lambda$$

Inverse Fourier Transform

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df$$

where $\omega = 2\pi f_0$, so can replace $2\pi f_0$ with ω. **BUT IN THIS FUCKING COURSE, THE TA SAID WE SHOULD USE $2\pi f_0$...**

Properties of the Fourier Transform

To simplify the calculation of Fourier transforms, we often use shortcuts based on the following properties:

1. Superposition

$$x_1(t) + x_2(t) \leftrightarrow X_1(f) + X_2(f)$$

2. Time Delay

$$x(t - t_0) \leftrightarrow e^{-j2\pi f_0 t} X(f)$$

3. Scale Change

$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{f}{a}\right)$$

4. Duality

$$X(t) \leftrightarrow x(-f)$$

Think of a function in terms of its graph.

5. Frequency Translation

$$x(t)e^{j2\pi f_0 t} \leftrightarrow X(f-f_0)$$

6. Modulation

$$x(t)\cos(2\pi f_0 t) \leftrightarrow \frac{1}{2}[X(f-f_0)+X(f+f_0)]$$

7. Differentiation

$$\frac{d}{dt}x(t) \leftrightarrow j2\pi f X(f)$$

8. Convolution

Time domain:

$$x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)\,d\tau$$

Frequency domain:

$$x_1(t) * x_2(t) \leftrightarrow X_1(f)X_2(f)$$

9. Multiplication

$$x_1(t)x_2(t) \leftrightarrow X_1(f) * X_2(f)$$

10. Delta Function

$$\delta(t) \leftrightarrow 1$$

11. Complex Exponential

$$e^{j2\pi f_0 t} \leftrightarrow \delta(f-f_0)$$

12. Area Property

The area under a function in one domain equals the value of the function at zero in the other domain.

13. Sinc Function

$$\text{rect}(\frac{t}{T}) \leftrightarrow T\,\text{sinc}(Tf)$$

Signal Bandwith

- **Strict bandwidth**: for signals whose spectrum restricted to some band, e.g. the sinc pulse, or a sum of sinusoids.
- **- 3 dB bandwidth**: The range of frequencies over which the energy spectral density (or power spectral density for power signals) is within 3 dB of the maximum value. This is used mostly in electronic circuit design, e.g. amplifier design.
- **x% bandwidth (e.g. x=99)**. The minimum range of frequencies over which the area under the energy spectral density curve (or power spectral density curve) equals x % of the total area.
- **rms bandwidth**

$$B_{rms} = (\frac{\int_{-\infty}^{\infty} f^2|X(f)|^2 df}{\int_{-\infty}^{\infty} |X(f)|^2 df})^{\frac{1}{2}}\ \text{Hz}$$

Time Duration of a Signal

$$T_{rms} = (\frac{\int_{-\infty}^{\infty} t^2|x(t)|^2 dt}{\int_{-\infty}^{\infty} |x(t)|^2 dt})^{\frac{1}{2}}$$

For any signal with rms time duratin T_{rms} and rms bandwidth B_{rms} we have

$$T_{rms}B_{rms} \geq \frac{1}{4\pi}$$

Hilbert Transform

General Form from Google is

$$H[f](x) = \frac{1}{\pi}\mathrm{PV}\int_{-\infty}^{\infty}\frac{f(y)}{x-y}\,dy$$

$$\begin{tikzcd}
x(t) \arrow[r, "\text{Fourier Transform}"] \arrow[d, "\text{Hilbert Transform}"] & X(f) \arrow[d, "-j\,\mathrm{sgn}(f)"] \\
\hat{x}(t) & -j\,\mathrm{sgn}(f)X(f) \arrow[l, "\text{Inverse Fourier Transform}"']
\end{tikzcd}$$

Hilbert transform of $x(t)$ is

$$\hat{x}(t) = \sin(2\pi f_0 t)$$

Signum function

$$\mathrm{sgn}(x) = \frac{x}{|x|}, \quad \text{for } x \neq 0$$

or

$$\mathrm{sgn}(x) = \begin{cases} -1, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ 1, & \text{if } x > 0 \end{cases}$$

Linear Time Invariant Systems

$$y(t) = h(t) * x(t)$$
$$Y(f) = H(f)X(f)$$

Amplitude Response: $A(f) = |H(f)|$

Phase Response: $\theta(f) = \arg(H(f))$

Energy Spectral Density of a Signal

Energy Spectra:

$$\psi_y(f) = |H(f)|^2\psi_x(f)$$